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Course Code: CSE325

AI & DATA SCIENCE LABORATORY

Course Objectives

This course will help the learner to design and implement techniques for creating simple autonomous agents capable of making effective decisions in fully informed and partially observable environments. Using appropriate algorithms to solve given AI problems. This course also helps the learner to employ T, Z, Anova test on data and derive inference using predictive analytics.

List of Experiments

1. Implement basic search strategies for selected AI applications
2. Implement A* and memory bounded A* algorithms
3. Implement genetic algorithms for AI tasks
4. Implement simulated annealing algorithms for AI tasks
5. Implement alpha-beta tree search
6. Implement backtracking algorithms for CSP
7. Implement Random Sampling
8. Demonstrating Z-test
9. Demonstrating T-test
10. Demonstrating ANOVA
11. Demonstrating Linear Regression
12. Demonstrating Logistic Regression

COURSE LEARNING OUTCOMES

Upon successful completion of this course, the learner will be able to:

CO No.	Course Outcome	Knowledge Level
1	Design and implement different techniques to develop simple autonomous agents that make effective decisions in fully informed, and partially observable settings.	K3
2	Apply appropriate algorithms for solving given AI problems.	K3
3	Design and implement logical reasoning agents and agents that can reason under uncertainty.	K3
4	Demonstrate the understanding of data distribution with samples	K3
5	Implement T-Test, Anova and Z-Test on sample data sets	K3
6	Employ Mathematical models in real world problems.	K3